Six Activations in Twelve Users

AP Statistics · Unit 3 · Topic 3.3 · B: 2026-12-03 · E: 2026-12-04 · TEACHER KEY

CED TETHER

- **VAR-1.H** Identify questions suggested by variation in the shapes of distributions of samples taken from the same population. **[Skill 1.A]**
- **VAR-1.H.1** Variation in shapes of data distributions may be random or not.

Essential question: When should a simulated sampling distribution be trusted more than a normal approximation?

DOK-3 skill: 4.B · **FRQ pattern:** `normal-approximation-vs-simulation-for-a-skewed-tail`

DOK rationale: Part (c) coordinates an expected-count check, the simulated distribution's shape, and two tail estimates to choose a defensible method, bound the evidence, and explain the consequence of the rejected analysis.

First take → rules + (a)–(c) → turn in **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Let students choose between the two reported tail probabilities without confirming a method.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board slide at minute 33. Circulate and require a shape or expected-count check before a probability choice.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose the normal-model or simulation estimate and cite one visible feature.
- Use the rules box to work (a) and (b); keep the chance explanation tied to the stated model.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What does one bar in this simulated distribution count?
- Which expected count warns you that a symmetric curve may miss the tail?
- Why do simulated samples with more than six activations still count?
- Can the two methods disagree in size yet lead to the same 5 percent decision?

Adult role

Solo teacher. Circulate 5 pairs. Ask for the model, shape check, tail event, and reader consequence; do not accept a method choice based only on which probability is smaller.

[WARN] LOOK FOR

- Treating every sampling distribution as normal regardless of expected counts.
- Counting exactly six activations instead of six or more.
- Saying a small chance probability proves why the activation rate differs.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Chooses the simulation using failed expected counts 2.4 and 9.6 or the right-skewed shape.
- **judgment-2** (required) — Uses $4/200{=}0.020$ below 0.05 as evidence against the model, without claiming proof.
- **judgment-3** (required) — Explains that 0.005 makes the result seem four times rarer and overstates the evidence.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

Common mistakes

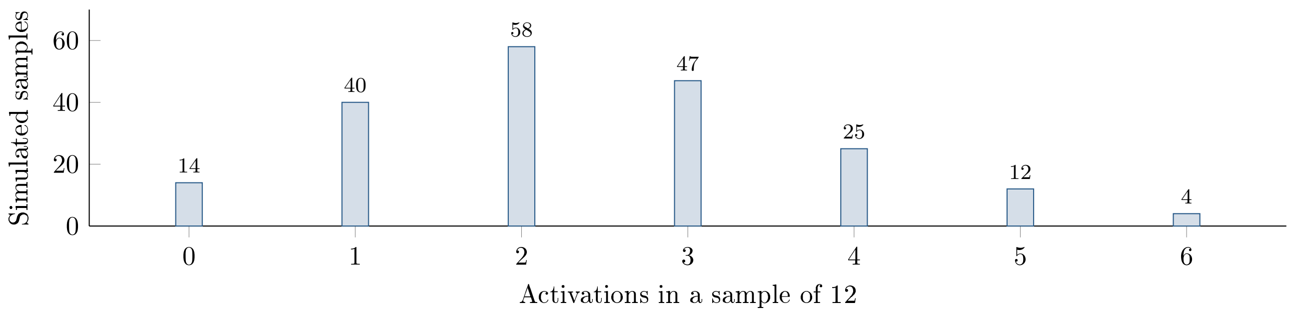
- Using $6/12 = 0.50$ as the chance probability
- Counting only the bar at 6 without recognizing that all larger results would also be at least as extreme
- Checking observed successes and failures instead of expected counts under p_0 for this model
- Calling the two percent estimate proof that the developer intentionally misstated the rate

[STAR] TODAY'S PROBLEM — annotated

Suppose a developer's model says $p_0 = 0.20$ of new app users activate voice commands. An SRS of 12 users includes 6 activations, so $\hat{p} = 0.50$. The chart shows 200 simulated samples of 12 independent users under this model.

Mara: "A normal model for $\hat{p}$ gives $P(\hat{p} \geq 0.50) \approx 0.005$, so that is the chance probability I would report."

Jules: "The simulated distribution is not very normal. I would estimate the chance probability directly from its tail."



FIRST TAKE (5 min)

Whose method would you trust more, Mara's smooth curve or Jules's repeated trials? Give one reason from the picture.
Key: Not graded. Preserve the normal-model versus matched-simulation tension.

Rules callout "CHECK THE SHAPE BEFORE USING NORMAL (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** State the most common number of activations in the simulated samples and describe the distribution as roughly symmetric, skewed left, or skewed right.

Key: Two activations is most common (58 simulations). The distribution is skewed right.

DOK 2 **(b)** Use the chart to estimate $P(\hat{p} \geq 0.50)$ under $p_0 = 0.20$. Then calculate np_0 and $n(1 - p_0)$ and assess the expected-count condition for a normal approximation.

Key: The simulated tail is $4/200 = 0.020$. Expected counts $12(0.20) = 2.4$ and $12(0.80) = 9.6$ both fall below 10, so the normal condition fails.

DOK 3 **(c)** In 3–4 sentences, choose a method using the simulated tail and either the shape or expected counts. Apply the 5% evidence rule. Explain how reporting 0.005 instead of 0.020 would change a reader's impression.

Key: Use Jules's simulation: the failed expected counts and right skew weaken the normal approximation. The tail $0.020 < 0.05$ gives evidence against the model, without proving a cause. Reporting 0.005 makes the result seem four times rarer and overstates the evidence.

EXIT

One thing the rules box changed about my first take: ______________________________